Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

Code Quiz: An AI-Driven Platform for Adaptive Code Assessment and Gamified Leaning

Fionn McGoldrick | Atlantic Technological University (ATU Galway)

GitHub: https://github.com/fionnmcgoldrick123/programming-quiz-app

Introduction & Motivation

LLM-generated quizzes with real-time ML difficulty prediction and topic tagging.

Motivation

- Education cost
- Static platforms
- Unstructured AI output
- Personalised learning
- Difficult platforms

Tech Stack

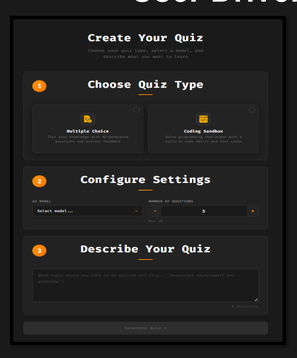
Frontend	Backend	Database	Machine Learning
React	FastAPI	PostgreSQL	scikit-learn
Vite	Python	Supabase	NumPy
TS			Pandas

Infrastructure

Docker, Nginx

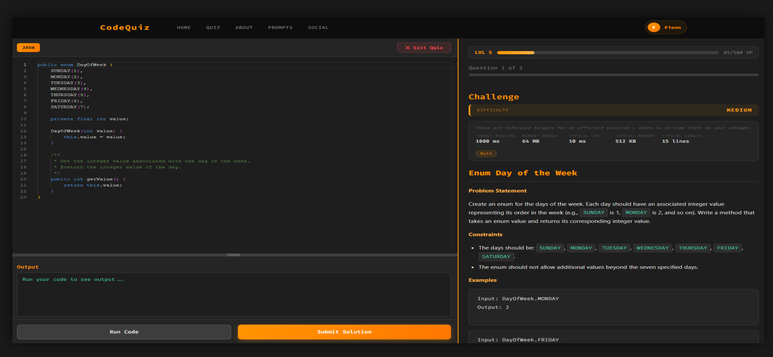
AI Quiz Generation

User Driven AI Quiz Configuration



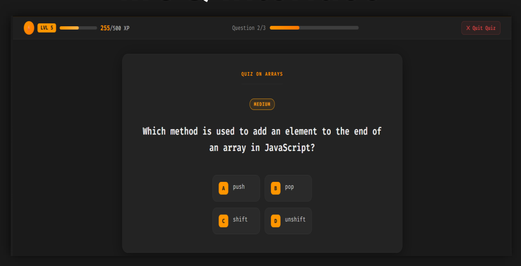
- MCQ + Coding
- AI Models: GPT-4o, Llama
- Question Limit: 20
- Topic Prompt
- Generate on Submit

Coding Sandbox Interface



Monaco Editor (syntax highlighting)	ML difficulty badge	Runtime, memory, CPU, length
Topic tag badge	Markdown problem + examples	Run | Submit (test cases)

MCQ Interface



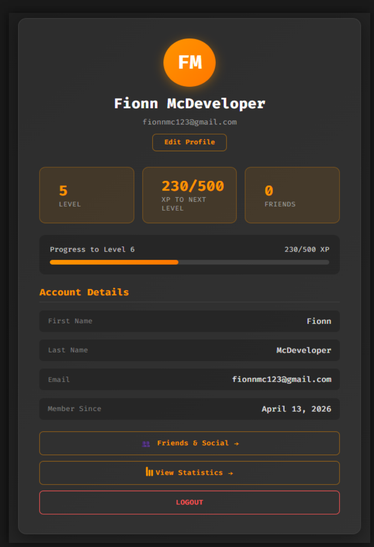
- ML predicted difficulty badge per question
- Four answer options A, B, C, D buttons
- progress tracker, top of screen

Performance Tracking and Statistics

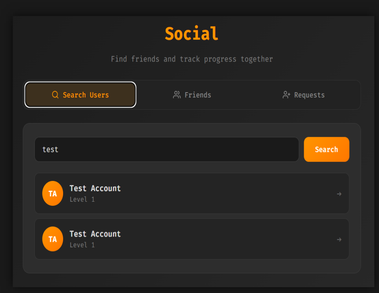
App Tracks User ...

- Strongest Topics
- Accuracy
- History
- Programming Preferences

Gamification & Social



- Personal user accounts
- Persistent User-Specific Data Displayed
- User Account Customisation
- Level & Current XP Visible



- Add Friends
- View Accounts
- Search Users

Difficulty Classifier

Training Data: IBM CodeNet Dataset

- 4,000+ problems
- Acceptance rate → difficulty
- Balanced 3 classes

Features

- 27 features + TF-IDF
- CPU, memory, code size
- Text stats: words, sentences
- Keyword flags (DP, graphs, etc)
- Word + char n-grams

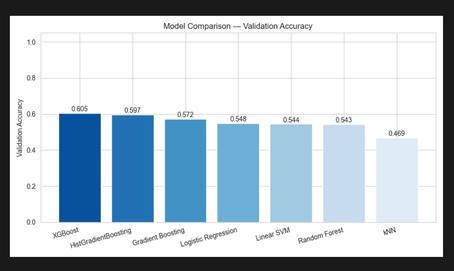
Preprocessing

- Numeric branch: median imputation then standard scaling
- Text branch: FeatureUnion of word and char TF-IDF vectorisers

Training

60	20	20
Train	Validation	Test

7 models benchmarked with RandomizedSearchCV, cv=5, scoring on F1-macro



XGBoost

Topic Tag Classifier

Training Data: Leetcode Dataset

- LeetCode problem dataset, title plus problem description plus topic tags per row
- Tags appearing in fewer than 60 problems excluded, leaving 13 classes

Array | String | Math | Hash Table | Table | Dynamic Programming | Binary Search | Divide & Conquer | Sliding Window | Graph | Greedy | Linked Lists | Recursion

Features

- Title + description combined
- HTML removed (regex)
- TF-IDF (word 1–3, char 3–5)
- Word n-grams → phrases
- Char n-grams → code tokens

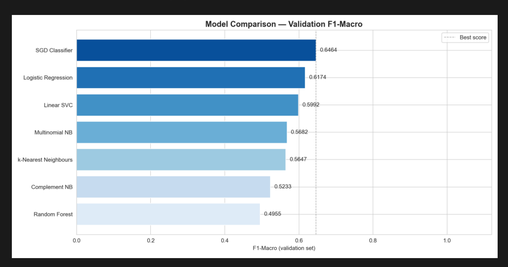
Preprocessing

- FeatureUnion concatenates word and char TF-IDF outputs into one feature matrix
- Sublinear TF scaling and English stop word removal applied to word vectoriser

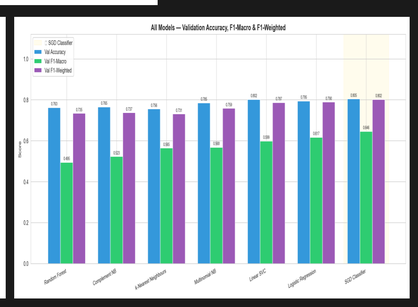
Training

60	20	20
Train	Validation	Test

All models wrapped in OneVsRestClassifier for multi-label support



SGD Classifier



Future Works

- Open Source
- Improve Models
- User billing system

Conclusion

Achieved:

- Full stack AI quiz platform
- Live ML inference
- Difficulty from 27 features
- Topic tagging (13 classes)
- Code execution (5 languages)
- XP, social, persistence

References

- IBM Research. *Project CodeNet: A Large-Scale AI for Code Dataset.* 2021.
- OpenAI. *GPT-4o-mini–Technical Report.* openai.com
- Meta AI. *Llama 3.1: Open Foundation Models.* ai.meta.com
- Pedregosa et al. *Scikit-learn: Machine Learning in Python.* JMLR, 2011.
- Chen & Guestrin. *XGBoost: A Scalable Tree Boosting System.* KDD, 2016.
- Microsoft. *Monaco Editor.* github.com/microsoft/monaco-editor

Acknowledgements

A special thank you to my project supervisor Douglas Mota Dias, and to ATU for their support and resources.